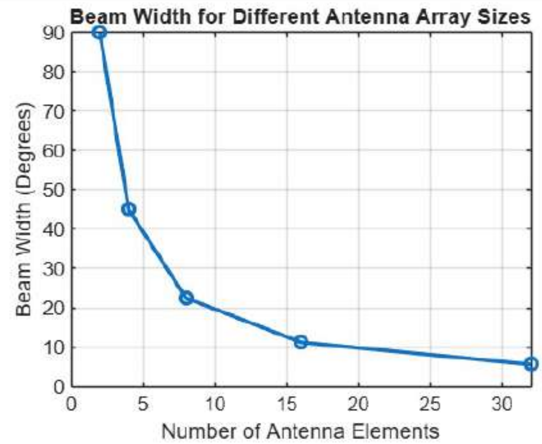
exp 5 object 1.mlx × | exp 5 object 2.mlx × | exp 5 object 3.mlx × | exp 6 object 1.mlx × | exp 6 object 2.mlx × | exp 6 object 3.mlx × | exp 7 object 1.mlx × | exp 7 object 2.mlx * ×

/MATLAB Drive/exp 7 object 2.mlx

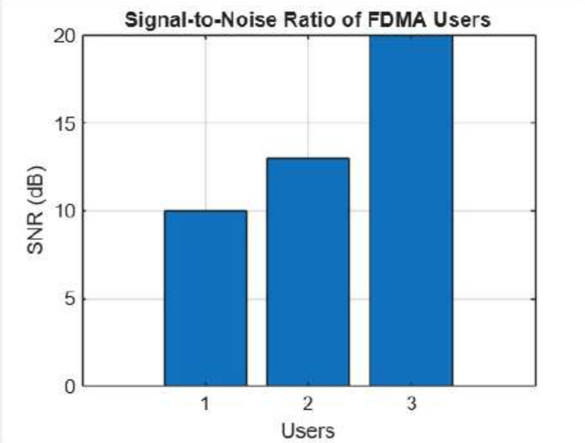
```
clc;
clear;
close all;
antennas = [2 4 8 16 32]; % Number of Antenna Elements
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)
figure
plot(antennas, beamwidth,'o-','LineWidth',2)
grid on

xlabel('Number of Antenna Elements')
ylabel('Beam Width (Degrees)')
title('Beam Width for Different Antenna Array Sizes')
```



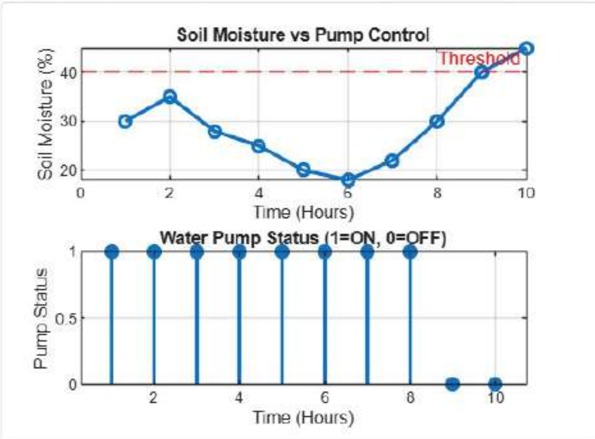
exp2t1.m × exp3t1.m × exp4t1.m × exp4t2.m × exp 1 object 1.mlx × exp 2 object 2.mlx × exp 3 object 3.mlx ×

/MATLAB Drive/exp 3 object 3.mlx

```
clc; clear; close all;
Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)
Users = 1:3;
SNR = 10*log10(Ps./Pn); % SNR in dB
disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on
```



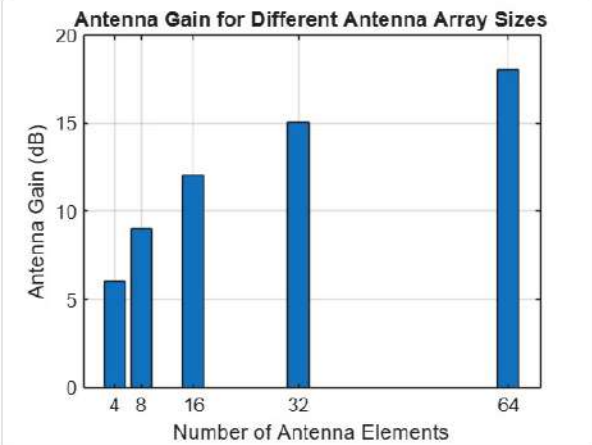
```
clc; clear; close all;
% Simulated soil moisture (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;

% Pump control logic (1 = ON, 0 = OFF)
pump_status = soil_moisture < threshold;
figure
% -------- Graph 1: Soil Moisture vs Pump Status --------
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2)
title('Soil Moisture vs Pump Control')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
```



exp 7 object 2.mlx × | exp 7 object 3.mlx × | exp 8 object 1.mlx × | exp 8 object 2.mlx × | exp 8 object 3.mlx × | exp 9 object 1.mlx × | exp 9 object 2.mlx × | exp 9 object 3.mlx * ×

/MATLAB Drive/exp 9 object 3.mlx

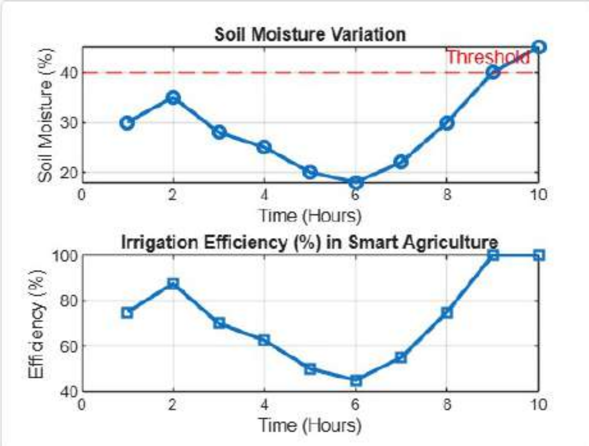
```matlab
clc;
clear;
close all;
antennas = [4 8 16 32 64]; % Number of Antenna Elements
gain = 10*log10(antennas); % Antenna Gain (dB)
figure
bar(antennas, gain)
grid on
xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('Antenna Gain for Different Antenna Array Sizes')
```



9 object 2.mlx × | exp 9 object 3.mlx × | exp 10 object 1.mlx × | exp 10 object 2.mlx × | exp 10 object 3.mlx × | exp 11 object 1.mlx × | exp 11 object 2.mlx × | exp 11 object 3.mlx * ×

/MATLAB Drive/exp 11 object 3.mlx

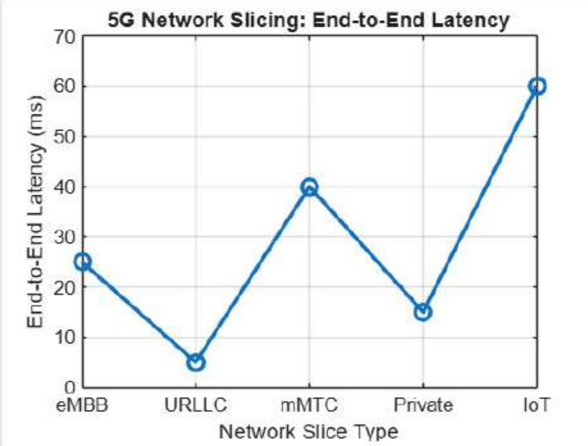
```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;
% Irrigation efficiency calculation (%)
efficiency = (soil_moisture ./ threshold) * 100;
efficiency(efficiency > 100) = 100; % limit to 100%
figure
% -------- Graph 1: Soil Moisture vs Efficiency --------
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2)
title('Soil Moisture Variation')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
```



exp 8 object 2.mlx × | exp 8 object 3.mlx × | exp 9 object 1.mlx × | exp 9 object 2.mlx × | exp 9 object 3.mlx × | exp 10 object 1.mlx × | exp 10 object 2.mlx × | exp 10 object 3.mlx * ×

/MATLAB Drive/exp 10 object 3.mlx

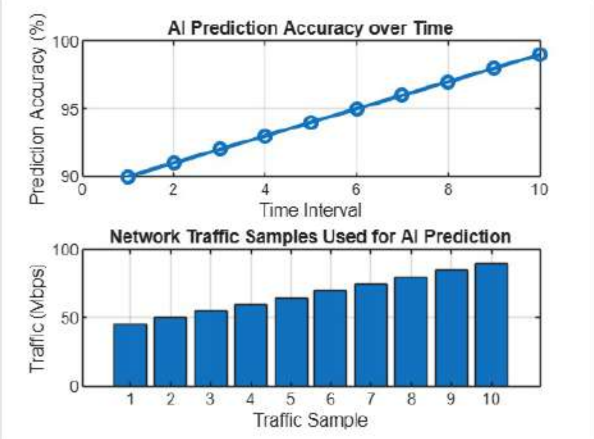
```
clc;
clear;
close all;
slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
grid on;

xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');
ylim([0 70]);
```



2 object 3.mlx × | exp 13 object 1.mlx × | exp 13 object 2.mlx × | exp 13 object 3.mlx × | exp 14 object 1.mlx × | exp 14 object 2.mlx × | exp 14 object 3.mlx × | exp 15 object 1.mlx * ×

/MATLAB Drive/exp 15 object 1.mlx

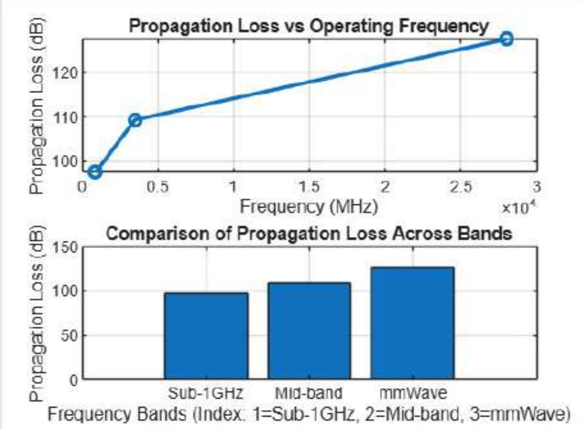
```
clc;
clear;
close all;
% Time samples
t = 1:10;
% Prediction Accuracy (%)
Accuracy = [90 91 92 93 94 95 96 97 98 99];

% Actual Network Traffic (Mbps)
Traffic = [45 50 55 60 65 70 75 80 85 90];
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(t,Accuracy,'-o','LineWidth',2)
title('AI Prediction Accuracy over Time')
xlabel('Time Interval')
```



exp 1 object 1.mlx × | exp 1 object 2.mlx × | exp 1 object 3.mlx × | exp 2 object 1.mlx × | exp 2 object 2.mlx × | exp 2 object 3.mlx × | exp 3 object 1.mlx × | exp 3 object 2.mlx * ×

/MATLAB Drive/exp 3 object 2.mlx

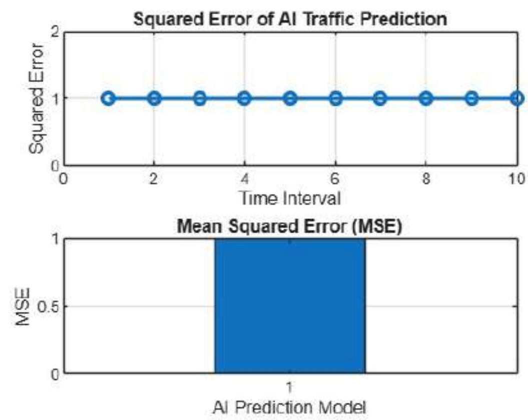
```
clc; clear; close all;
d = 2; % fixed distance (km)

f = [900 3500 28000]; % MHz
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
% -------- Graph 1: Frequency vs Loss --------
subplot(2,1,1)
plot(f,PL,'-o','LineWidth',2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% -------- Graph 2: Band Comparison --------
subplot(2,1,2)
```



3 object 1.mlx × exp 13 object 2.mlx × exp 13 object 3.mlx × exp 14 object 1.mlx × exp 14 object 2.mlx × exp 14 object 3.mlx × exp 15 object 1.mlx × exp 15 object 2.mlx * ×

/MATLAB Drive/exp 15 object 2.mlx

```
clc;
clear;
close all;
% Time intervals
t = 1:10;
% Actual and Predicted Traffic (Mbps)
Actual = [50 55 60 62 68 70 75 80 85 90];
Predicted = [49 54 61 63 67 71 74 81 84 89];
% Squared Error
SqError = (Actual - Predicted).^2;
% Mean Squared Error
MSE = mean(SqError);
disp(['Mean Squared Error (MSE) = ', num2str(MSE)])
figure
% -------- Graph 1 --------
subplot(2,1,1)
```

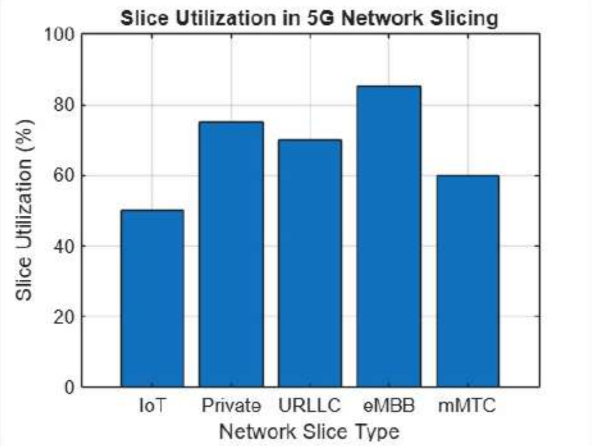


Command Window

exp 8 object 1.mlx × | exp 8 object 2.mlx × | exp 8 object 3.mlx × | exp 9 object 1.mlx × | exp 9 object 2.mlx × | exp 9 object 3.mlx × | exp 10 object 1.mlx × | exp 10 object 2.mlx ×

/MATLAB Drive/exp 10 object 2.mlx

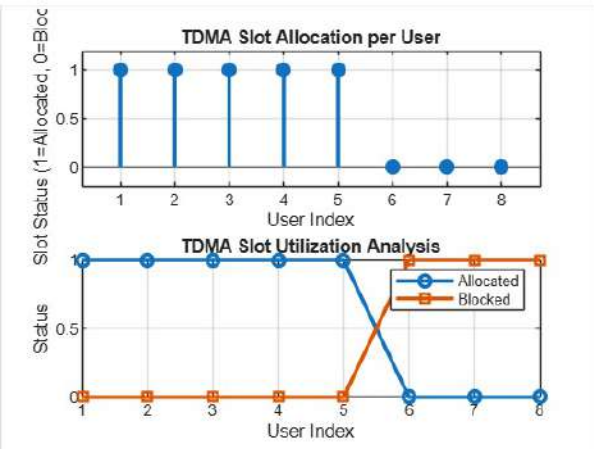
```
clc;
clear;
close all;
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
utilization = [85 70 60 75 50];

figure;
bar(slices, utilization);
grid on;
xlabel('Network Slice Type');
ylabel('Slice Utilization (%)');
title('Slice Utilization in 5G Network Slicing');
ylim([0 100]);
```



exp 1 object 1.mlx × | exp 1 object 2.mlx × | exp 1 object 3.mlx × | exp 2 object 1.mlx × | exp 2 object 2.mlx × | exp 2 object 3.mlx × | +

/MATLAB Drive/exp 2 object 3.mlx

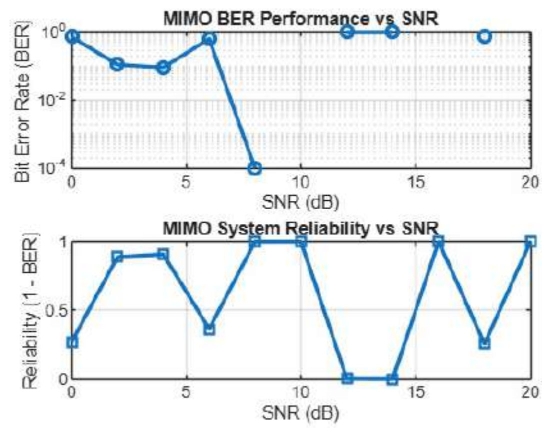
```
clc; clear; close all;
Users = 1:8; % Total users
TotalSlots = 5; % Available slots
Alloc = zeros(1,8); % Initialize allocation
Alloc(1:TotalSlots) = 1; % Assign slots to first users
Blocked = 1 - Alloc; % Remaining users blocked
disp('User Wise Slot Allocation')
disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Bl
figure
% Graph 1: Allocation per user
subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
```



exp 3 object 3.mlx × exp 4 object 1.mlx × exp 4 object 2.mlx × exp 4 object 3.mlx × exp 5 object 1.mlx * ×

/MATLAB Drive/exp 5 object 1.mlx

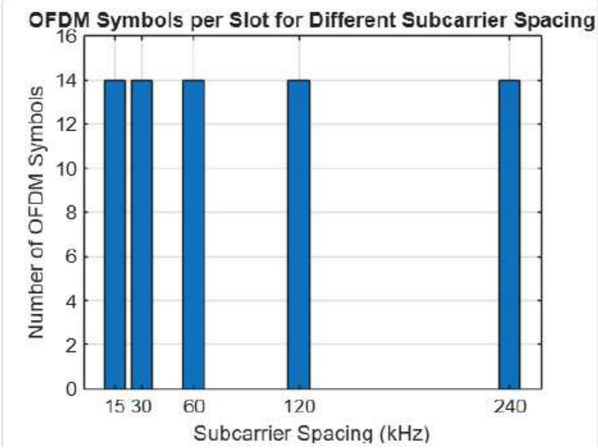
```
clc; clear; close all;
% SNR range
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2; Nr = 2;
BER = zeros(1,length(SNR));

for i = 1:length(SNR)
    bits = randi([0 1],10000,1);
    symbols = 2*bits-1; % BPSK
    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
    noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
    tx = H(1,1)*symbols' + noise(1,:);
    rx = real(tx) > 0;
    BER(i) = sum(rx' ~= bits)/length(bits);
```



exp 4 object 2.mlx × | exp 4 object 3.mlx × | exp 5 object 1.mlx × | exp 5 object 2.mlx × | exp 5 object 3.mlx × | exp 6 object 1.mlx × | exp 6 object 2.mlx × | exp 6 object 3.mlx * ×

/MATLAB Drive/exp 6 object 3.mlx

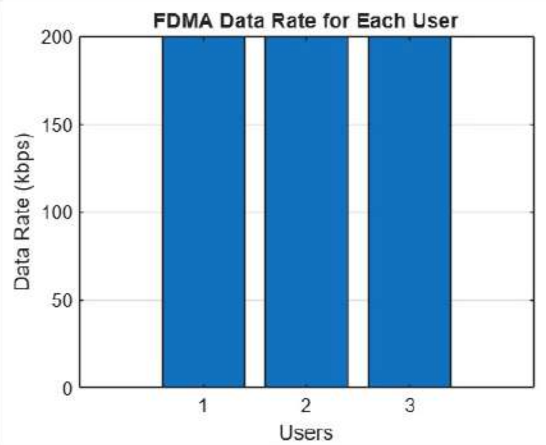
```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
figure
bar(scs, symbols)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
ylim([0 16])
```



exp2t1.m | exp3t1.m | exp4t1.m | exp4t2.m | exp 1 object 1.mlx | exp 2 object 2.mlx

/MATLAB Drive/exp 2 object 2.mlx

```
clc; clear; close all;
BW = 200; % Channel Bandwidth (kHz)
Users = 3;
M = 2; % BPSK Modulation
DataRate = BW*log2(M); % Data Rate (kbps)
Rate = DataRate*ones(1,Users);
disp(' User Data Rate (kbps)')
disp([(1:Users)' Rate'])
figure
bar(1:Users,Rate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User')
grid on
```

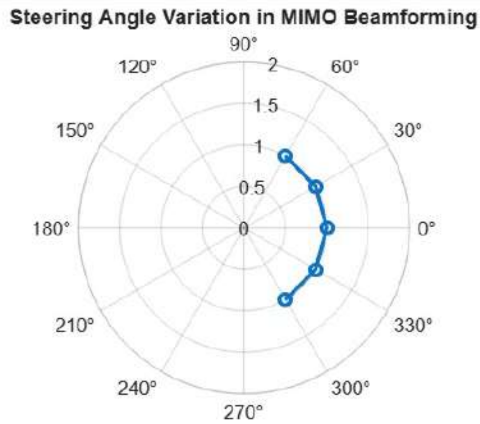


Command Window

exp 5 object 3.mlx × | exp 6 object 1.mlx × | exp 6 object 2.mlx × | exp 6 object 3.mlx × | exp 7 object 1.mlx × | exp 7 object 2.mlx × | exp 7 object 3.mlx × | exp 8 object 1.mlx * ×

/MATLAB Drive/exp 8 object 1.mlx

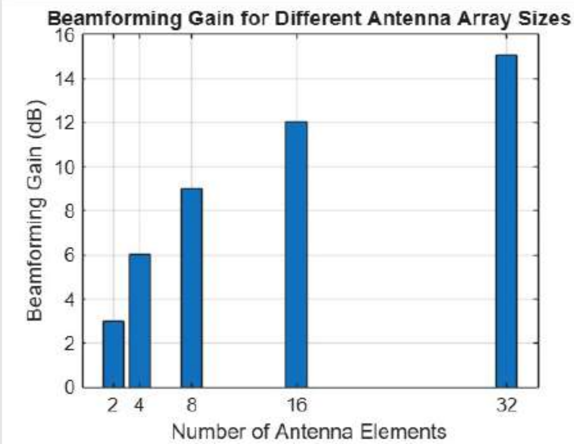
```
clc;
clear;
close all;
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
gain = ones(size(angle)); % Normalized Gain
figure
polarplot(deg2rad(angle), gain,'o-','LineWidth',2)
title('Steering Angle Variation in MIMO Beamforming')
```



exp 4 object 3.mlx × exp 5 object 1.mlx × exp 5 object 2.mlx × exp 5 object 3.mlx × exp 6 object 1.mlx × exp 6 object 2.mlx × exp 6 object 3.mlx × exp 7 object 1.mlx * ×

/MATLAB Drive/exp 7 object 1.mlx

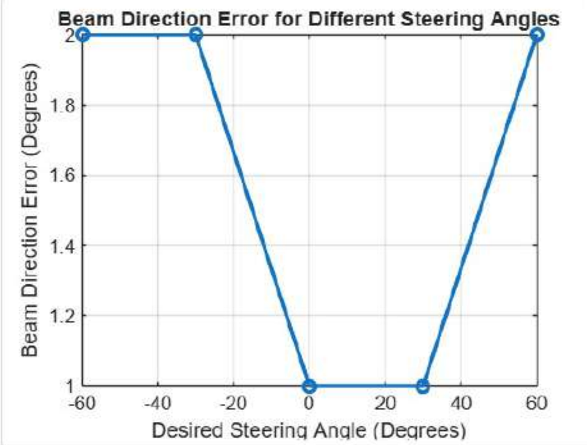
```matlab
clc;
clear;
close all;
antennas = [2 4 8 16 32]; % Number of Antenna Elements
gain = 10*log10(antennas); % Beamforming Gain (dB)
figure
bar(antennas, gain)
grid on

xlabel('Number of Antenna Elements')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Antenna Array Sizes')
```



exp 6 object 2.mlx × exp 6 object 3.mlx × exp 7 object 1.mlx × exp 7 object 2.mlx × exp 7 object 3.mlx × exp 8 object 1.mlx × exp 8 object 2.mlx × exp 8 object 3.mlx * ×

/MATLAB Drive/exp 8 object 3.mlx

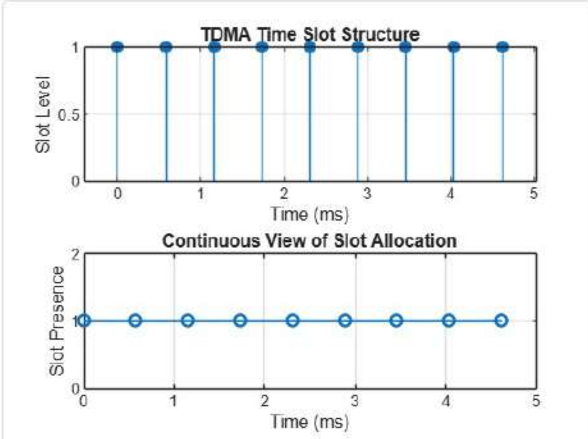
```
clc;
clear;
close all;
desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees)
actual = [-58 -32 1 29 62]; % Actual Beam Direction (degrees)
error = abs(desired - actual); % Beam Direction Error
figure
plot(desired, error, 'o-', 'LineWidth', 2)
grid on
xlabel('Desired Steering Angle (Degrees)')
ylabel('Beam Direction Error (Degrees)')
title('Beam Direction Error for Different Steering Angles')
```



exp 1 object 1.mlx × | exp 2 object 2.mlx × | exp 3 object 3.mlx × | exp 2 object 1.mlx ×

/MATLAB Drive/exp 2 object 1.mlx

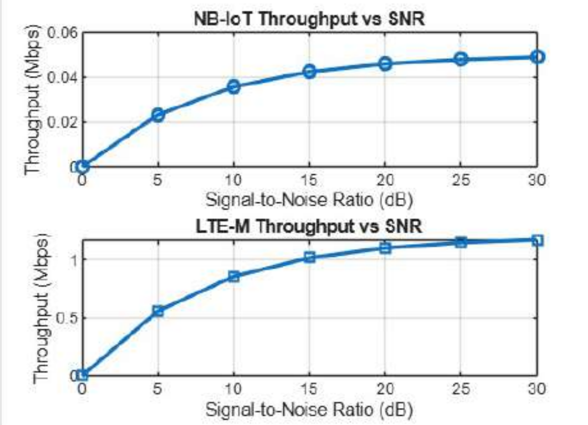
```
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;
t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')

xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```



9 object 3.mlx × | exp 10 object 1.mlx × | exp 10 object 2.mlx × | exp 10 object 3.mlx × | exp 11 object 1.mlx × | exp 11 object 2.mlx × | exp 11 object 3.mlx × | exp 12 object 1.mlx * ×

/MATLAB Drive/exp 12 object 1.mlx

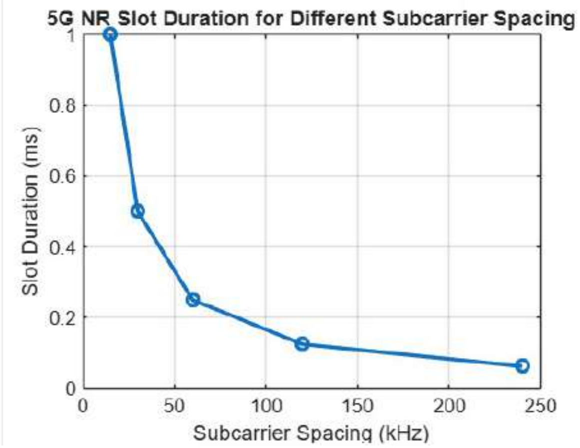
```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M = 1.2*(1-exp(-SNR/8));

disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
disp([SNR' NB_IoT' LTE_M'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Throughput vs SNR')
```



exp 4 object 1.mlx × | exp 4 object 2.mlx × | exp 4 object 3.mlx × | exp 5 object 1.mlx × | exp 5 object 2.mlx × | exp 5 object 3.mlx × | exp 6 object 1.mlx × | exp 6 object 2.mlx * ×

/MATLAB Drive/exp 6 object 2.mlx

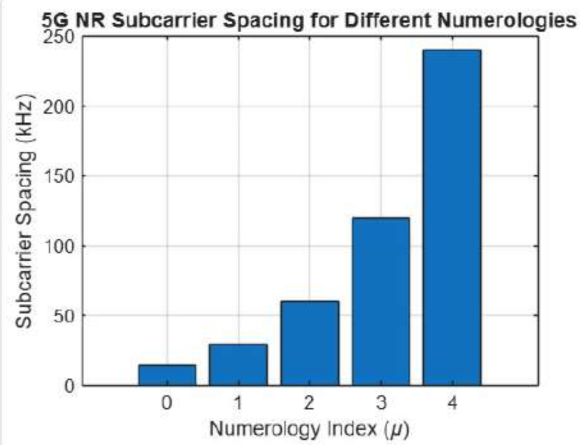
```
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)
plot(scs, slot, 'o-', 'LineWidth', 2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')
```



exp 3 object 3.mlx × exp 4 object 1.mlx × exp 4 object 2.mlx × exp 4 object 3.mlx × exp 5 object 1.mlx × exp 5 object 2.mlx × exp 5 object 3.mlx × exp 6 object 1.mlx ×

/MATLAB Drive/exp 6 object 1.mlx

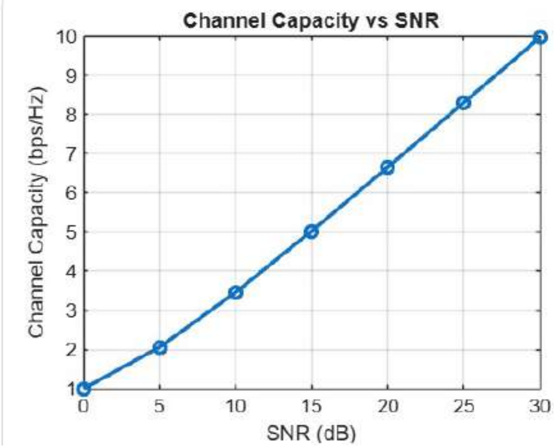
```
clc;
clear;
close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```



exp 7 object 1.mlx × | exp 7 object 2.mlx × | exp 7 object 3.mlx × | exp 8 object 1.mlx × | exp 8 object 2.mlx × | exp 8 object 3.mlx × | exp 9 object 1.mlx × | exp 9 object 2.mlx * ×

/MATLAB Drive/exp 9 object 2.mlx

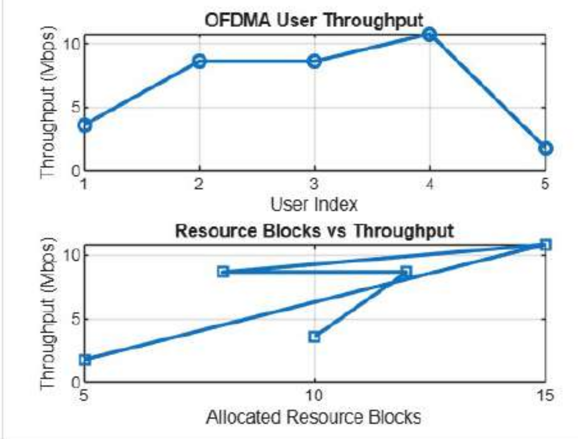
```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);

figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```



exp 3 object 3.mlx × | exp 4 object 1.mlx × | exp 4 object 2.mlx × | exp 4 object 3.mlx × | exp 5 object 1.mlx × | exp 5 object 2.mlx × | exp 5 object 3.mlx * × | +

/MATLAB Drive/exp 5 object 3.mlx

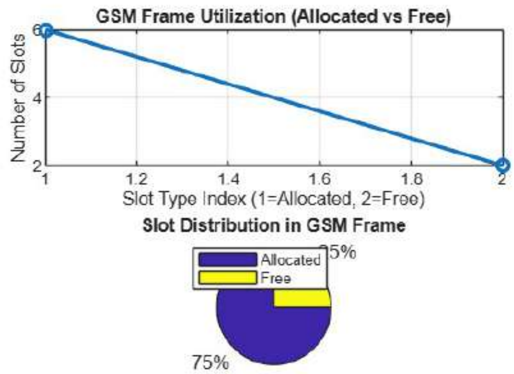
```
clc; clear; close all;
% Resource Block parameters
rb_bw = 180e3; % 180 kHz per RB
% Users
users = 1:5;
% Allocated Resource Blocks per user
alloc_RB = [10 12 8 15 5];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

% Display results
disp('User Throughput (Mbps):')
disp(throughput')
figure
```



exp 1 object 1.mlx | exp 1 object 2.mlx | exp 1 object 3.mlx | exp 2 object 1.mlx | exp 2 object 2.mlx *

/MATLAB Drive/exp 2 object 2.mlx

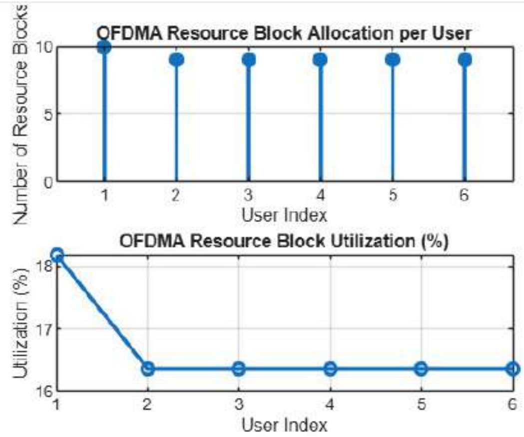
```
FreeSlots = TotalSlots - AllocatedSlots;
Utilization = (AllocatedSlots/TotalSlots)*100;
disp('Frame Utilization (%)')
disp(Utilization)
figure
% Graph 1: Simple Line Plot (SAFE - no bar function)
subplot(2,1,1)
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
ylabel('Number of Slots')
grid on

% Graph 2: Pie Chart (SAFE)
subplot(2,1,2)
pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')
```



exp 3 object 3.mlx × exp 4 object 1.mlx ×

/MATLAB Drive/exp 4 object 1.mlx

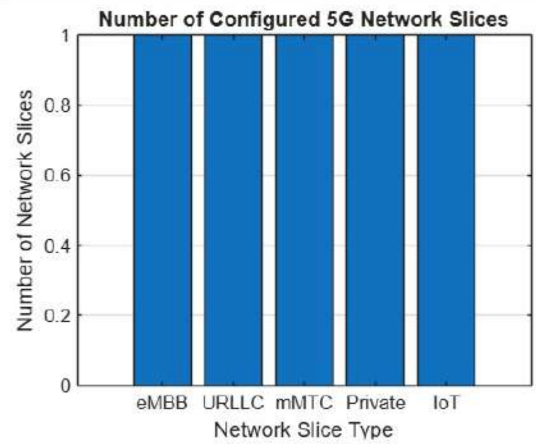
```
clc; clear; close all;
% System parameters
bandwidth = 10e6; % 10 MHz
rb_size = 180e3; % 180 kHz per RB
% Total Resource Blocks
total_RB = floor(bandwidth / rb_size);
% Users
users = 1:6;
% RB allocation (simple equal + remainder distribution)
base = floor(total_RB / length(users));

alloc = base * ones(1,length(users));
alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users)
utilization = (alloc / total_RB) * 100;
disp('Total Resource Blocks:')
disp(total_RB)
```



exp 7 object 3.mlx × exp 8 object 1.mlx × exp 8 object 2.mlx × exp 8 object 3.mlx × exp 9 object 1.mlx × exp 9 object 2.mlx × exp 9 object 3.mlx × exp 10 object 1.mlx * ×

/MATLAB Drive/exp 10 object 1.mlx

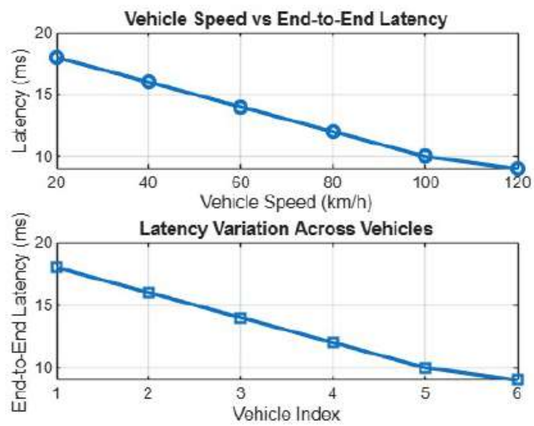
```
clc;
clear;
close all;
slices = {'eMBB','URLLC','mMTC','Private','IoT'};

count = [1 1 1 1 1]; % One instance of each network slice
figure
bar(count)
grid on
set(gca,'XTickLabel',slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')
```



) object 3.mlx × | exp 11 object 1.mlx × | exp 11 object 2.mlx × | exp 11 object 3.mlx × | exp 12 object 1.mlx × | exp 12 object 2.mlx × | exp 12 object 3.mlx × | exp 13 object 1.mlx * ×

/MATLAB Drive/exp 13 object 1.mlx

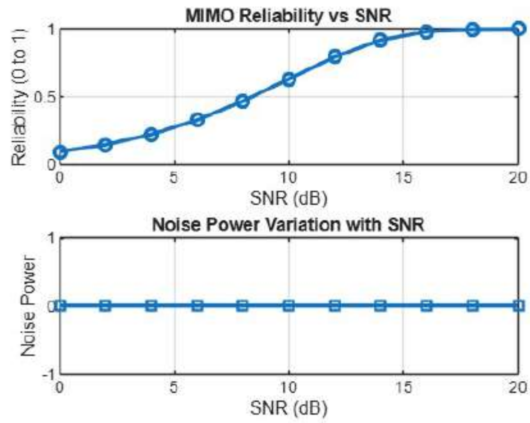
```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])

figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
```



exp 3 object 3.mlx × | exp 4 object 1.mlx × | exp 4 object 2.mlx × | exp 4 object 3.mlx × | exp 5 object 1.mlx × | exp 5 object 2.mlx * ×

/MATLAB Drive/exp 5 object 2.mlx

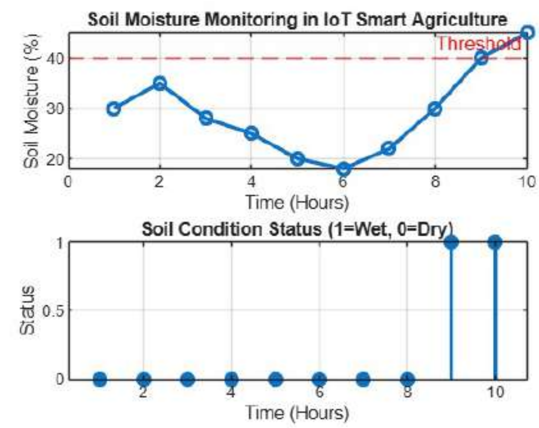
```
clc; clear; close all;
% SNR range (dB)
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2; Nr = 2;
% Simulated reliability metric (inverse BER model)
Reliability = zeros(1,length(SNR));
for i = 1:length(SNR)
    % simple channel model effect
    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
    noise_power = 1/SNR(i);
    % reliability increases with SNR
    Reliability(i) = 1 - exp(-SNR(i)/10);
end
```



xp 8 object 3.mlx × | exp 9 object 1.mlx × | exp 9 object 2.mlx × | exp 9 object 3.mlx × | exp 10 object 1.mlx × | exp 10 object 2.mlx × | exp 10 object 3.mlx × | exp 11 object 1.mlx * ×

/MATLAB Drive/exp 11 object 1.mlx

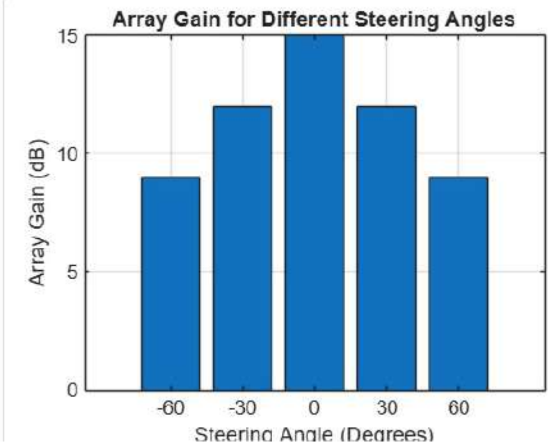
```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;
figure
% -------- Graph 1: Soil Moisture Variation --------
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Monitoring in IoT Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')

grid on
hold on
```



exp 6 object 1.mlx × | exp 6 object 2.mlx × | exp 6 object 3.mlx × | exp 7 object 1.mlx × | exp 7 object 2.mlx × | exp 7 object 3.mlx × | exp 8 object 1.mlx × | exp 8 object 2.mlx * ×

/MATLAB Drive/exp 8 object 2.mlx

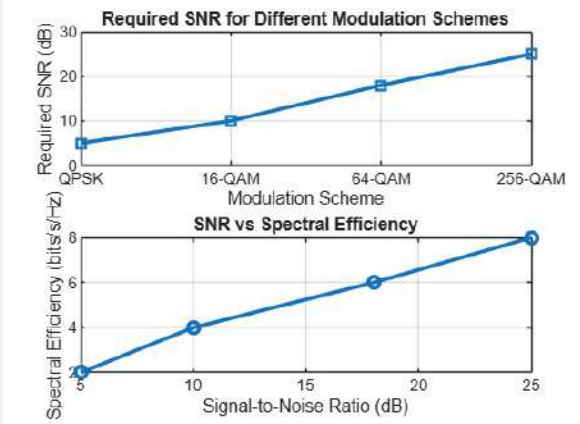
```
clc;
clear;
close all;
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
gain = [9 12 15 12 9]; % Array Gain (dB)
figure
bar(angle, gain)
grid on
xlabel('Steering Angle (Degrees)')
ylabel('Array Gain (dB)')
title('Array Gain for Different Steering Angles')
```



2 object 2.mlx × exp 12 object 3.mlx × exp 13 object 1.mlx × exp 13 object 2.mlx × exp 13 object 3.mlx × exp 14 object 1.mlx × exp 14 object 2.mlx × exp 14 object 3.mlx * ×

/MATLAB Drive/exp 14 object 3.mlx

```
clc;
clear;
close all;
% Modulation schemes
Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};
% Required SNR (dB)
SNR = [5 10 18 25];
% Spectral Efficiency (bits/s/Hz)
Efficiency = [2 4 6 8];
disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation',SNR',Efficiency', ...
    'VariableNames',{'Modulation','SNR_dB','Efficiency'}))
figure
% -------- Graph 1 --------
subplot(2,1,2)

```

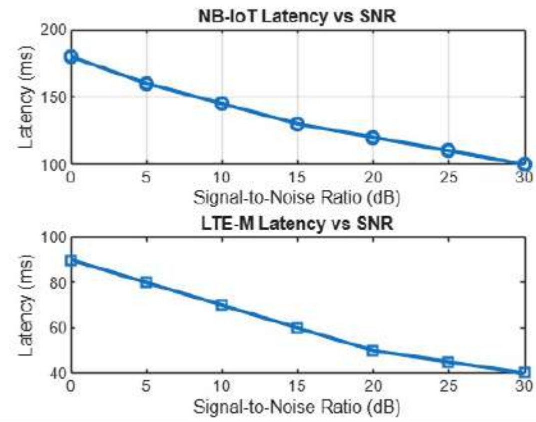


Command Window

0 object 1.mlx × | exp 10 object 2.mlx × | exp 10 object 3.mlx × | exp 11 object 1.mlx × | exp 11 object 2.mlx × | exp 11 object 3.mlx × | exp 12 object 1.mlx × | exp 12 object 2.mlx * ×

/MATLAB Drive/exp 12 object 2.mlx

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Latency (ms)
NB_IoT = [180 160 145 130 120 110 100];
LTE_M = [90 80 70 60 50 45 40];
disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')
disp([SNR' NB_IoT' LTE_M'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
```

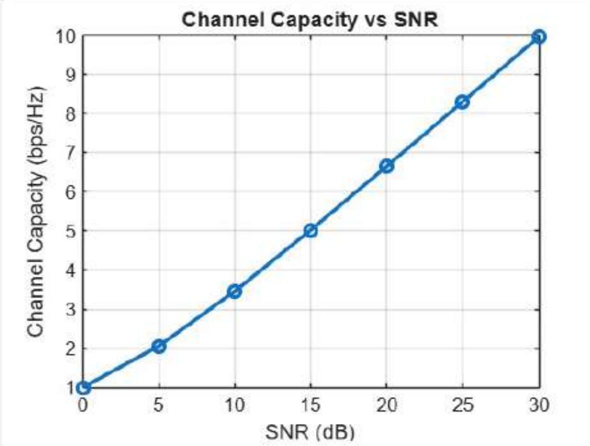


Command Window

exp 6 object 3.mlx × exp 7 object 1.mlx × exp 7 object 2.mlx × exp 7 object 3.mlx × exp 8 object 1.mlx × exp 8 object 2.mlx × exp 8 object 3.mlx × exp 9 object 1.mlx * ×

/MATLAB Drive/exp 9 object 1.mlx

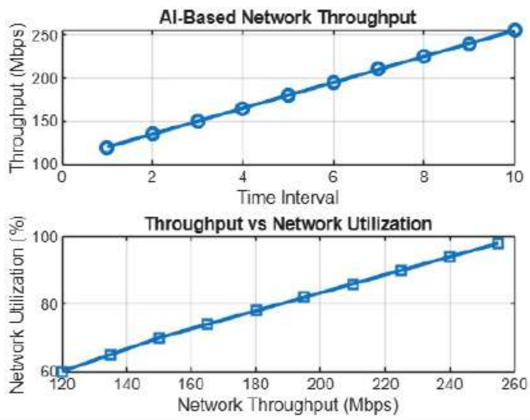
```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```



3 object 2.mlx × | exp 13 object 3.mlx × | exp 14 object 1.mlx × | exp 14 object 2.mlx × | exp 14 object 3.mlx × | exp 15 object 1.mlx × | exp 15 object 2.mlx × | exp 15 object 3.mlx ×

/MATLAB Drive/exp 15 object 3.mlx

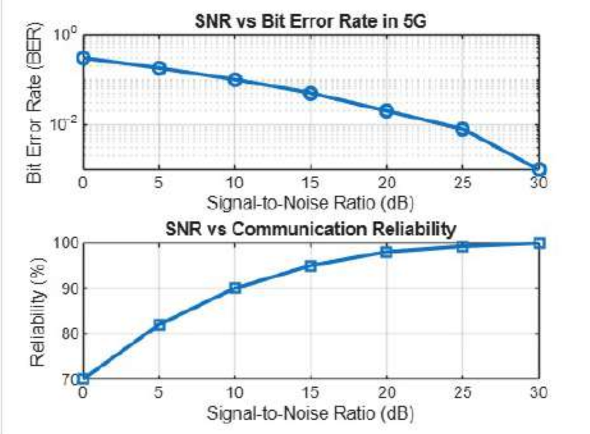
```
clc;
clear;
close all;
% Time intervals
t = 1:10;
% Network Throughput (Mbps)
Throughput = [120 135 150 165 180 195 210 225 240 255];
% Network Utilization (%)
Utilization = [60 65 70 74 78 82 86 90 94 98];
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(t,Throughput,'-o','LineWidth',2)
title('AI-Based Network Throughput')
xlabel('Time Interval')
ylabel('Throughput (Mbps)')
```



2 object 1.mlx × | exp 12 object 2.mlx × | exp 12 object 3.mlx × | exp 13 object 1.mlx × | exp 13 object 2.mlx × | exp 13 object 3.mlx × | exp 14 object 1.mlx × | exp 14 object 2.mlx * ×

/MATLAB Drive/exp 14 object 2.mlx

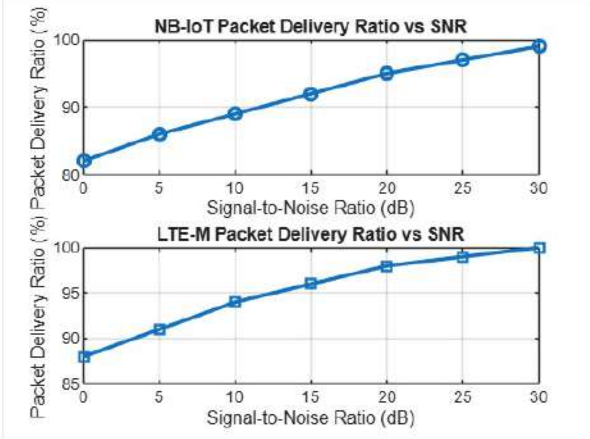
```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Bit Error Rate (BER)
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
% Reliability (%)
Reliability = (1-BER)*100;
disp('SNR(dB) BER Reliability(%)')
disp([SNR' BER' Reliability'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
semilogy(SNR,BER,'-o','LineWidth',2)
title('SNR vs Bit Error Rate in 5G')
```



0 object 2.mlx × | exp 10 object 3.mlx × | exp 11 object 1.mlx × | exp 11 object 2.mlx × | exp 11 object 3.mlx × | exp 12 object 1.mlx × | exp 12 object 2.mlx × | exp 12 object 3.mlx * ×

/MATLAB Drive/exp 12 object 3.mlx

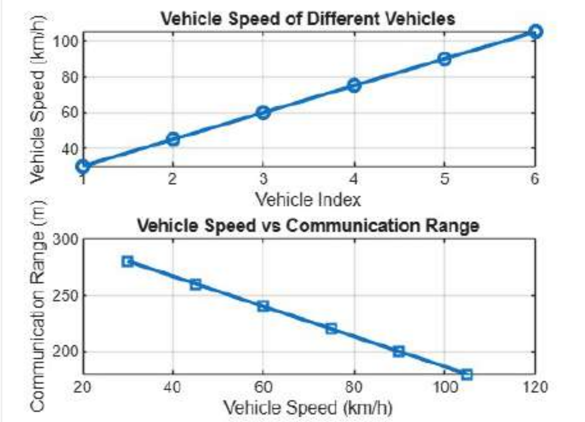
```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Packet Delivery Ratio (%)
NB_IoT = [82 86 89 92 95 97 99];
LTE_M = [88 91 94 96 98 99 100];
disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%)')
disp([SNR' NB_IoT' LTE_M'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
```



1 object 2.mlx × exp 11 object 3.mlx × exp 12 object 1.mlx × exp 12 object 2.mlx × exp 12 object 3.mlx × exp 13 object 1.mlx × exp 13 object 2.mlx × exp 13 object 3.mlx * ×

/MATLAB Drive/exp 13 object 3.mlx

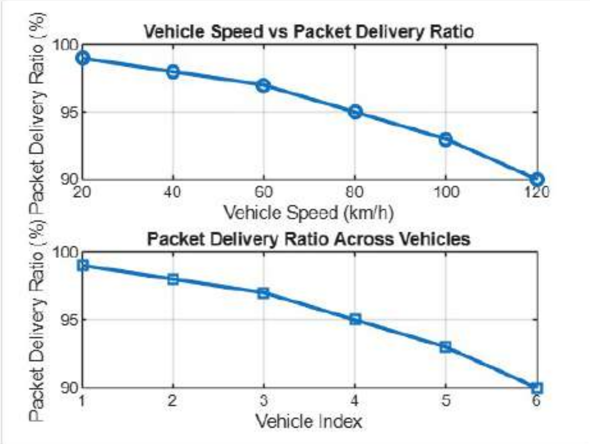
```
clc;
clear;
close all;
% Vehicle IDs
vehicle = 1:6;
% Vehicle Speed (km/h)
speed = [30 45 60 75 90 105];
% Communication Range (m)
range = [280 260 240 220 200 180];
disp('Vehicle Speed(km/h) Communication Range(m)')
disp([vehicle' speed' range'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(vehicle,speed,'-o','LineWidth',2)
title('Vehicle Speed of Different Vehicles')
```



object 1.mlx | exp 11 object 2.mlx | exp 11 object 3.mlx | exp 12 object 1.mlx | exp 12 object 2.mlx | exp 12 object 3.mlx | exp 13 object 1.mlx | exp 13 object 2.mlx *

/MATLAB Drive/exp 13 object 2.mlx

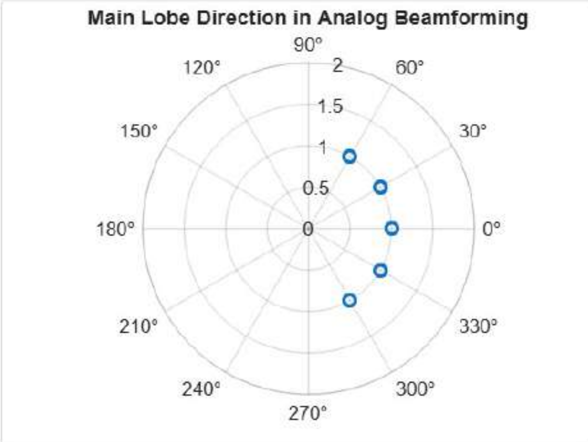
```
clc;
clear;
close all;
% Vehicle Speed (km/h)

speed = 20:20:120;
% Simulated Packet Delivery Ratio (%)
PDR = [99 98 97 95 93 90];
disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')
disp([speed' PDR'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(speed,PDR,'-o','LineWidth',2)
title('Vehicle Speed vs Packet Delivery Ratio')
xlabel('Vehicle Speed (km/h)')
```



exp 5 object 2.mlx × exp 5 object 3.mlx × exp 6 object 1.mlx × exp 6 object 2.mlx × exp 6 object 3.mlx × exp 7 object 1.mlx × exp 7 object 2.mlx × exp 7 object 3.mlx * ×

/MATLAB Drive/exp 7 object 3.mlx

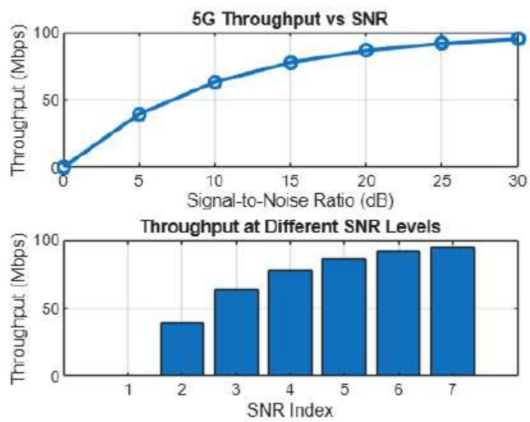
```
clc;
clear;
close all;
theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
gain = [1 1 1 1 1]; % Constant normalized gain
figure
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
title('Main Lobe Direction in Analog Beamforming')
```



object 3.mlx × | exp 12 object 1.mlx × | exp 12 object 2.mlx × | exp 12 object 3.mlx × | exp 13 object 1.mlx × | exp 13 object 2.mlx × | exp 13 object 3.mlx × | exp 14 object 1.mlx * ×

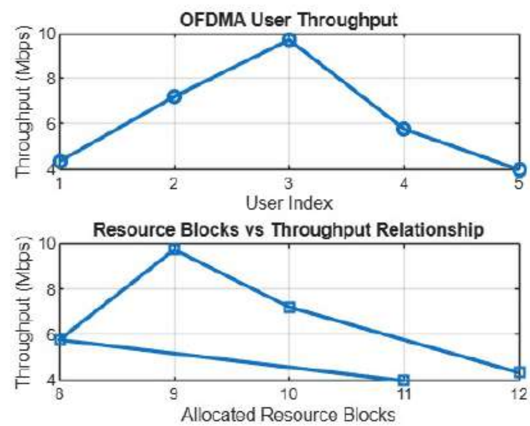
/MATLAB Drive/exp 14 object 1.mlx

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation

Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,Throughput,'-o','LineWidth',2)
```



/MATLAB Drive/exp 4 object 3.mlx

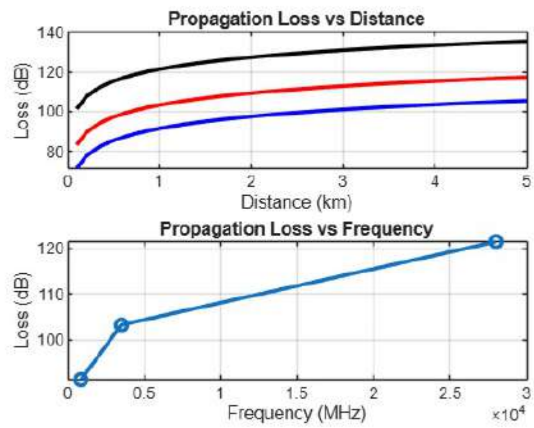
```matlab
clc; clear; close all;
% System parameters
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
users = 1:5;
% Allocated Resource Blocks per user
alloc_RB = [12 10 9 8 11];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')
disp(throughput')
figure
% -------- Graph 1: Throughput per User --------
subplot(2,1,1)

```



exp 1 object 1.mlx × | exp 1 object 2.mlx × | exp 1 object 3.mlx × | exp 2 object 1.mlx × | exp 2 object 2.mlx × | exp 2 object 3.mlx × | exp 3 object 1.mlx * ×

/MATLAB Drive/exp 3 object 1.mlx

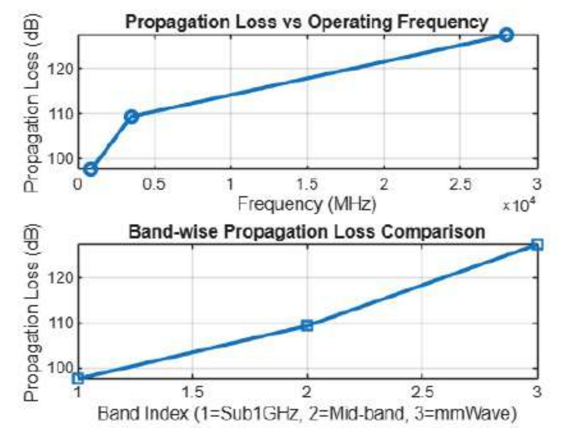
```
clc; clear; close all;
d = 0.1:0.1:5; % Distance (km)
f = [900 3500 28000]; % Frequency (MHz)
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
% -------- Graph 1 --------
subplot(2,1,1)

plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
```



exp 1 object 2.mlx | exp 1 object 3.mlx | exp 2 object 1.mlx | exp 2 object 2.mlx | exp 2 object 3.mlx | exp 3 object 1.mlx | exp 3 object 2.mlx | exp 3 object 3.mlx

/MATLAB Drive/exp 3 object 3.mlx

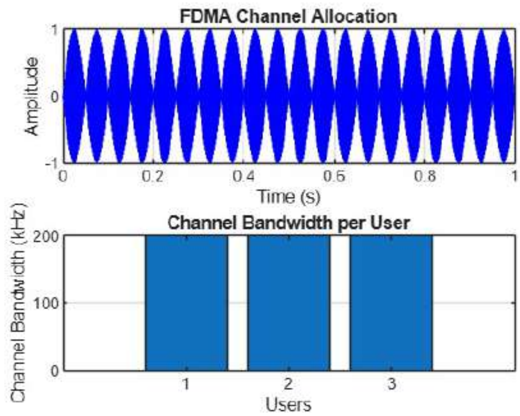
```
clc; clear; close all;
% Frequency bands (MHz)
f = [900 3500 28000];
% Distance (km)
d = 2;
% Propagation Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);
figure
% -------- Graph 1: Simple Line Plot (SAFE) --------
subplot(2,1,1)
plot(f, PL, '-o', 'LineWidth', 2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% -------- Graph 2: Safe Comparison Plot --------
```



exp2t1.m × exp3t1.m × exp4t1.m × exp4t2.m × exp 1 object 1.mlx * ×

/MATLAB Drive/exp 1 object 1.mlx

```
clc; clear; close all;
fs=1000; t=0:1/fs:1;
TBW=600; Users=3;

CBW=TBW/Users;
fc=[100 300 500];
m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on
xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidt
```



Command Window